

Iterative Development of NeuroDegenRX: From Proof-of-Concept to Production-Ready Framework

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Abstract

This report chronicles the stepwise, iterative development of NeuroDegenRX, a mechanism-driven computational framework for neurodegenerative drug triage. Starting from a minimal proof-of-concept focused on CNS drug-likeness scoring, we progressively incorporated mechanistic toxicity filtering, pathway perturbation modeling, and stochastic disease progression simulation. Each iteration addressed specific challenges—interpretability, false positive rates, pathway crosstalk, and uncertainty propagation—through rigorous testing and integration of domain knowledge. The final platform integrates all modules into a transparent, auditable system that decomposes every output into its biological components. This report details the technical evolution, validation steps, and key citations that guided our decisions.

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1 Introduction

The discovery of disease-modifying therapies for neurodegenerative disorders such as Alzheimer’s disease (AD) remains one of the most challenging and resource-intensive endeavors in pharmaceutical research (1). Traditional computational approaches often rely on black-box models that produce aggregate scores or rankings without providing insight into the underlying biological reasoning. This opacity hampers scientific trust and impedes medicinal chemistry optimization (2).

NeuroDegenRX was conceived to address this gap by building a transparent, mechanism-grounded decision-support system. Rather than a single monolithic release, the platform evolved through a series of targeted iterations, each designed to validate a core capability and incorporate user feedback. This document describes that journey—from a simple Python script calculating CNS multiparameter optimization (MPO) scores to a fully integrated web application with disease simulation and explainability layers.

2 Iteration 1: CNS MPO Proof-of-Concept

2.1 Objective

The primary goal of the first iteration was to establish a baseline capability: assess whether a compound possesses physicochemical properties conducive to central nervous system (CNS) penetration. We aimed to reproduce and extend the CNS MPO scoring system pioneered by Pfizer (3; 4).

2.2 Implementation

We implemented a Python script that, given a compound’s SMILES string, computes six key parameters:

- Molecular Weight (MW)
- Calculated LogP (clogP)
- Hydrogen Bond Donors (HBD)
- Topological Polar Surface Area (TPSA)
- Strongest Basic pKa (pKa)
- Number of Rotatable Bonds (RotB)

Each parameter was normalized to a score between 0 and 1 using piecewise linear functions derived from (author?) (3). The total CNS MPO score was the sum of these six components.

2.3 Validation

We tested the script on a curated set of 150 marketed CNS drugs and 150 non-CNS drugs (from the ChEMBL database (5)). The ROC-AUC for distinguishing CNS from non-CNS compounds was 0.87, consistent with literature values.

2.4 Limitations and Lessons

While the scoring worked well, early user interviews with medicinal chemists revealed a critical shortcoming: *the score alone was not actionable*. A compound scoring 3.8 versus 4.2 provided no guidance on which parameter to optimize. This feedback underscored the need for *decomposability*—a principle that would guide all subsequent iterations.

3 Iteration 2: Mechanistic Toxicity Filtering

3.1 Objective

The second iteration aimed to identify potential safety liabilities early in the triage process. We sought to flag structural alerts, off-target interactions, and metabolic risks.

3.2 Implementation

We integrated three toxicity screening layers:

1. **Structural alerts:** Implemented using the SMARTS patterns from the PAINS filter (6) and the Brenk alert set (7).
2. **Target-based risk:** Compiled a list of high-risk targets (e.g., hERG, 5-HT2B, CYP enzymes) from literature and public databases (ChEMBL, DrugBank). We used precomputed similarity ensembles to predict off-target interactions (8).
3. **Metabolic interactions:** Flagged compounds likely to inhibit major CYP isoforms (CYP3A4, CYP2D6, CYP2C9) using rule-based models (9).

3.3 Challenges and Refinements

Initial testing on a set of 50 compounds with known clinical toxicity revealed an unacceptably high false positive rate—over 80% of compounds triggered at least one alert. This “alert fatigue” rendered the output useless for decision-making.

To address this, we introduced a *severity triage system*:

- **High priority:** Alerts with strong clinical evidence (e.g., hERG IC₅₀ $\leq$ 1 μ M, known hepatotoxins).

- **Moderate priority:** Alerts requiring follow-up (e.g., CYP inhibition with IC50 $\geq 10 \mu\text{M}$).
- **Low priority:** Structural alerts with uncertain relevance (e.g., PAINS in non-interfering chemotypes).

Additionally, we added *contextual explanations*: each alert included the exact substructure, the mechanism of toxicity (if known), and references to similar compounds that succeeded or failed.

3.4 Validation

Retrospective analysis on 30 drugs withdrawn due to toxicity (10) showed that our tiered system correctly flagged 90% of them as high priority, while reducing false positives on safe drugs to 25%.

4 Iteration 3: Pathway Perturbation Engine

4.1 Objective

The third iteration sought to connect molecular interactions to systems-level biological impact. We needed to translate a compound’s polypharmacology into pathway activity scores relevant to neurodegeneration.

4.2 Implementation

We curated a set of 12 pathways central to AD pathogenesis (neuroinflammation, autophagy, synaptic health, mitochondrial function, etc.) from KEGG (11) and Reactome (12). For each target, we assigned:

- A *pathway role* $R \in \{-1, +1\}$ indicating whether the target activates or inhibits the pathway.
- A *modality* $D \in \{-1, +1\}$ indicating whether the compound is an agonist or inhibitor.

Binding affinities (Ki or IC50, in nM) were transformed into weights using:

$$W = -\log_{10}(\text{Affinity})$$

The pathway score was then computed as:

$$\text{Pathway Score} = \sum (W \times D \times R)$$

4.3 Handling Pathway Crosstalk

Early versions produced scores that were difficult to interpret due to overlapping targets participating in multiple pathways. We addressed this by:

- Normalizing scores by the total possible weight in each pathway.
- Displaying confidence intervals based on the number of targets engaged.
- Providing a network visualization of pathway–target interactions (using Cytoscape.js in the frontend).

4.4 Validation

We tested the engine on compounds with known effects: rapamycin (autophagy activator) and lipopolysaccharide (neuroinflammation inducer). The pathway scores correctly reflected expected directions and magnitudes. We also compared predictions against transcriptomic data from the LINCS L1000 database (13) for a subset of compounds; correlation coefficients ranged from 0.6 to 0.8.

5 Iteration 4: Disease Progression Simulation

5.1 Objective

The ultimate goal was to simulate how a compound might alter the long-term trajectory of AD. This required a dynamic model incorporating multiple pathological hallmarks and their interactions.

5.2 Model Design

We adopted a systems of ordinary differential equations (ODEs) framework inspired by previous mechanistic models of AD (14; 15). Seven state variables were chosen:

$X = [\text{Amyloid}, \text{Tau}, \text{Neuroinflammation}, \text{Synaptic Health}, \text{Mitochondrial Function}, \text{Oxidative Stress}, \text{Autophagy}]$

Baseline progression was modeled as:

$$\frac{dX_i}{dt} = \alpha_i \cdot X_i \cdot (1 - X_i/K_i) + \sum_j \beta_{ij} X_j$$

where α_i are intrinsic growth rates, K_i carrying capacities, and β_{ij} cross-coupling terms derived from literature (16).

5.3 Drug Effect Integration

Drug-induced pathway scores (from Iteration 3) were mapped to modifications of the corresponding state variables:

$$\frac{dX_i}{dt} = \text{baseline}_i + \gamma_i \cdot \text{PathwayScore}_i$$

where γ_i scales the drug effect strength based on estimated brain exposure.

5.4 Stochastic Uncertainty

Recognizing that long-term projections are inherently uncertain, we introduced stochastic terms:

$$X_{t+1} = X_t + \Delta t \cdot \frac{dX}{dt} + \epsilon_t$$

with $\epsilon_t \sim \mathcal{N}(0, \sigma_t^2)$ and $\sigma_t^2 = \sigma_0^2(1 + \alpha t)$ to model increasing variance over time.

5.5 Calibration and Testing

We calibrated the baseline parameters using longitudinal data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) (17). The model reproduced typical progression patterns (e.g., amyloid plateau followed by tau acceleration) with mean absolute error $\leq 15\%$ over 5 years.

6 Iteration 5: Integration and Transparency

6.1 Objective

The final iteration aimed to unify all four modules into a single, accessible platform with a focus on interpretability and user experience. The backend needed to support explainability at every level.

6.2 System Architecture

We built the backend using Django 4.2 and Django REST Framework, with PostgreSQL as the primary database. Each computational engine operates as an independent service, communicating via REST APIs. The entire system is containerized with Docker to ensure reproducibility (18).

6.3 Explainability Layer

A dedicated explanation engine decomposes every output:

- **CNS MPO:** Individual parameter contributions are shown, with optimization hints.
- **Toxicity:** Each flagged issue includes a mechanistic rationale, severity, and mitigation suggestions.
- **Pathway scores:** Target-level contributions are listed, with confidence based on evidence quality.
- **Disease trajectories:** Uncertainty intervals and sensitivity analyses accompany projections.

6.4 Frontend Development

The React/TypeScript frontend (described in a companion report) implements these explanations through modular components: `UncertaintyIndicator`, `MechanismExplanation`, and interactive visualizations using Recharts and Cytoscape.js.

6.5 User Testing

We conducted a user study with 12 medicinal chemists and 8 pharmacologists. Participants were asked to prioritize three compounds using either the final platform or a baseline “black-box” dashboard. The NeuroDegenRX users showed 37% faster decision times and reported significantly higher confidence in their selections (4.2 vs. 2.8 on a 5-point scale). Qualitative feedback emphasized the value of “seeing why” a compound scored as it did.

7 Conclusion

The iterative development of NeuroDegenRX demonstrates that building trust in computational tools requires more than predictive accuracy—it demands mechanistic transparency, actionable explanations, and rigorous validation at each step. Starting from a simple MPO calculator, we progressively integrated toxicity filtering, pathway modeling, and disease simulation, addressing real-world challenges such as false positives, pathway crosstalk, and uncertainty propagation. The resulting platform provides researchers with a unified, auditable system for neurodegenerative drug triage.

Future work will focus on expanding the pathway knowledge base, incorporating additional disease models (e.g., Parkinson’s, ALS), and enabling community contributions to the underlying evidence graphs.

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